SOLUTION KEY

T.A.U. Engineering

	DIFFERENTIAL EQUATIONS Midterm II	
Course : *Differential Equations* Year : *2025-2026* Semester : *Fall* Lecturer : *Derya Altıntan* Date : *12 December 2025* Time : *15:00* Duration : *120 minutes*	Surname : Name : Number : Signature :	Department:
	7 QUESTION 100 POINTS	

1	2	3	4	5	6	7	TOTAL	

1.(20 pts)Inside a cell, the amount of a certain mRNA molecule $M(t)$ changes over time. The production of M is proportional to its current amount. The degradation of M is proportional to the square of its current amount.

1. **Using the information above and introducing any proportionality constants needed, derive a differential equation for $M(t)$ and obtain its general solution.**

$$\frac{dM}{dt} = k_1 M - k_2 M^2$$

$$\frac{dM}{k_1 M - k_2 M^2} = dt$$

$$\frac{1}{M(k_1 - k_2 M)} = \frac{A}{M} + \frac{B}{k_1 - k_2 M}$$

$$k_1 A = 1 \Rightarrow A = 1/k_1$$

$$M(-k_2 A + B) = 0 \Rightarrow B = k_2 A$$

$$B = \frac{k_2}{k_1}$$

$$dt = \frac{1}{k_1 M} + \frac{k_2}{k_1}\frac{1}{k_1 - k_2 M}$$

$$t + C = \frac{1}{k_1}\ln|M| - \frac{1}{k_1}\ln|k_1 - k_2 M|$$

$$t + C = \frac{1}{k_1}\ln\left|\frac{M}{k_1 - k_2 M}\right|$$

2. **Let $M(0) = M_0$. Obtain the particular solution of the initial value problem symbolically, giving an explicit formula for $M(t)$ in terms of your constants and M_0.**

$M(0) = M_0$

$$C = \frac{1}{k_1}\ln\left|\frac{M_0}{k_1 - k_2 M_0}\right|$$

$$t = \frac{1}{k_1}\left(\ln\frac{|M|}{|k_1 - k_2 M|} - \ln\frac{|M_0|}{|k_1 - k_2 M_0|}\right)$$

$$t = \frac{1}{k_1}\ln\left(\frac{|M|\,|k_1 - k_2 M_0|}{|k_1 - k_2 M|\,|M_0|}\right)$$

$k_1 = k \quad k_2 = l$

$$t + C = \frac{1}{k}\ln\left|\frac{y}{k - ly}\right| \quad e^{kt + kC} = \frac{y}{k - ly}$$

$$\frac{k - ly}{y} = e^{-kt}C$$

$$\frac{k}{y} - l = e^{-kt}C$$

$$y = \frac{k}{l}\left(e^{-kt}C + l\right)$$

2.(8+8 + 4 pts) Consider the spring mass system

$$u''(t) + \gamma u'(t) + 100\, u(t) = F(t), \quad u(0) = 2,\ u'(0) = 20 \qquad (*)$$

(a) Solve equation (*) for $\gamma = 0$, $F(t) = 0$. Find amplitude, period, and sketch the graph of the solution.

$u''(t) + 100u(t) = 0 \quad r^2 + 100 = 0 \quad r = \pm 10i$

$u(t) = A\cos 10t + B\sin 10t \quad u(0) = A = 2$

$u'(t) = -10A\sin 10t + B \cdot 10\cos 10t \quad u'(0) = 10B = 20 \quad B = 2$

$u(t) = 2\cos 10t + 2\sin 10t. \quad R = \sqrt{2^2+2^2} = 2\sqrt{2}$ amplitude $T = \frac{2\pi}{10}$ per

(b) Solve equation (*) for $\gamma = 10\sqrt{3}$, $F(t) = 0$. Find amplitude, period, and sketch the graph with their envelopes.

$u'' + 10\sqrt{3}u'(t) + 100u(t) = 0 \quad \Rightarrow r_{1,2} = \frac{-10\sqrt{3} \pm \sqrt{300-400}}{2} = -5\sqrt{3} \pm 5i$

$r^2 + 10\sqrt{3}r + 100 = 0$

$u(t) = e^{-5\sqrt{3}t}(A\cos 5t + B\sin 5t)$

$u'(t) = -5\sqrt{3}e^{-5\sqrt{3}t}(A\cos 5t + B\sin 5t) + e^{-5\sqrt{3}t}(-5A\sin 5t + 5B\cos 5t)$

$u(0) = A = 2$

$u'(0) = B = 4 + 2\sqrt{3} \quad -5\sqrt{3}A + 5B = 20 \Rightarrow 5B = 20 + 10\sqrt{3}$

$B = 4 + 2\sqrt{3}$

$u(t) = e^{-5\sqrt{3}t}(2\cos 5t + 20\sin 5t) \quad R = \sqrt{A^2+B^2} = 4\sqrt{2+\sqrt{3}}$

(c) Assume $\gamma = 0$, $F(t) = 5\sin 2w + \cos w$. For which values of w the system has a resonance.

$u'' + 100u(t) = 5\sin 2\omega t + \cos \omega$

Resonance $u(t) = A\cos 10t + B\sin 10t \quad \omega_0 = 10$

If $\omega = \omega_0$ or $2\omega = \omega_0$

we have resonance $\omega = 10$ or $\omega = 5$

$u(t) = R\cos(\omega t - \delta)e^{-\alpha t}$

$= 4\sqrt{2+\sqrt{3}}\cos(5t - \delta)e^{-5\sqrt{3}t}$

$A = R\cos\delta \Rightarrow \cos\delta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$

$B = R\sin\delta \Rightarrow \sin\delta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\tan\delta = \frac{B}{A} \quad \tan\delta = \frac{B}{A} = 1 \quad \delta = 45.$

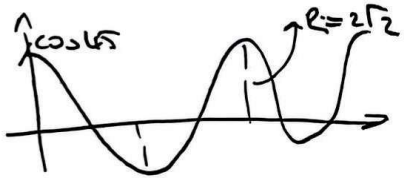
$T = \frac{2\pi}{10} = \pi/5.$



$R/m = \sqrt{100} = 10$

$u(t) = 2\sqrt{2}\cos(\omega_0 t - 45)$

$= 2\sqrt{2}\cos(10t - 45)$

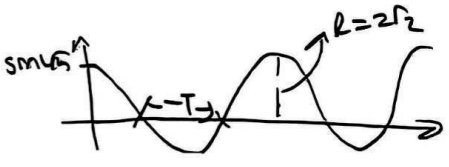


OR $A = R\sin\delta \Rightarrow \sin\delta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$

$B = R\cos\delta \Rightarrow \cos\delta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\tan\delta = \frac{A}{B} \quad \tan\delta = 45.$

$u(t) = 2\sqrt{2}\sin(10t + 45)$



8.(20 pts) Obtain the particular solution of the following initial value problem by using the variation of parameters method

$$t\left(y'' - 2y' + y\right) = 2e^t,\ t > 0,\ y(1) = 0,\ y'(1) = e.$$

$y'' - 2y' + y = \frac{2}{t} e^t \quad y(1)=0,\ y'(1)=e$

y_h $\quad x^2 - 2x + 1 = 0 \quad x = 1, 1 \quad y_h(t) = v_1 e^t + v_2 e^t \cdot t$

$(x-1)^2 = 0 \quad F.S.S = \{e^t, t\cdot e^t\}$

y_p $\quad v_1' e^t + v_2' (e^t \cdot t) = 0$

$v_1' e^t + v_2' (e^t + t\cdot e^t) = \frac{2}{t} e^t$ $\Bigg\}$ $w = \begin{vmatrix} e^t & te^t \\ e^t & (1+t)e^t \end{vmatrix} = e^{2t} \neq 0$

Or $\quad v_1' + v_2' = 0 \quad w = \begin{vmatrix} e^t & t \\ 1 & 1+t \end{vmatrix} = 1 \neq 0$

$v_1' + v_2' (1+t) = \frac{2}{t}$

$$v_1' = \begin{vmatrix} 0 & te^t \\ \frac{2}{t}e^t & e^t + te^t \end{vmatrix} \Big/ e^{2t} \qquad v_2' = \frac{\begin{vmatrix} e^t & 0 \\ e^t & \frac{2}{t}e^t \end{vmatrix}}{e^{2t}} = \frac{2}{t}$$

$v_1' = -2 \Rightarrow v_1(t) = -2t \qquad y_p(t) = -2t\cdot e^t + 2\ln|t| \cdot t \cdot e^t$

$v_2' = \frac{2}{t} \Rightarrow v_2(t) = 2\ln|t|$

$y_{c_1} = \underbrace{c_1 e^t + c_2 e^t \cdot t}_{y_h} \underbrace{- 2te^t + 2\ln|t| \cdot t \cdot e^t}_{y_p}$

$y_{c_1}(1) = (c_1 + c_2 - 2)e = 0 \Rightarrow c_1 + c_2 = 2$

$y_{c_1}' = c_1 e^t + c_2 (t+1)e^t - 2\left[e^t + te^t\right] + 2\cdot\frac{1}{t}\cdot t\cdot e^t$

$+ 2\ln|t|\left(e^t + t\cdot e^t\right) \qquad c_1 + 2c_2 = 3$

$c_1 = 1$

$y_{c_1}'(1) = c_1 \cdot e + 2c_2 e - 4e + 2e = e. \qquad c_2 = 1$

$y = e^t + e^t \cdot t - 2te^t + 2\ln|t|\, t\cdot e^t.$

4.(20 pts) Obtain the homogeneous differential equation whose particular solution is given as $y = 2x^2 + xe^x \cos x - 4 \sin 3x$ and write the general solution of obtained differential equation.

$F.S.S = \{1, x, x^2, e^x \cos x, e^x \sin x, xe^x \cos x, xe^x \sin x, \sin 3x, \cos 3x\}$

$x^2 \Rightarrow 0,0,0$

$xe^x \cos x \Rightarrow r = 1 \mp i;\ 1 \mp i$

$\sin 3x \Rightarrow r = \mp 3i$.

$L(D) = D^3 ((D-(1+i))(D-(1-i)))^2 (D^2+9)y$

$= D^3 (D^2 - 2D + 2)^2 (D^2+9) y$

$y(x) = c_1 + c_2 x + c_3 x^2 + c_4 e^x \cos x + c_5 e^x \sin x + c_6 x e^x \cos x$

$+ c_7 x e^x \sin x + c_8 \sin 3x + c_9 \cos 3x$

8, Consider the matrix

$$A = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$$

(a) Find P and J for A.

$\det(A-\lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = -\lambda(2-\lambda)+1 = \lambda^2-2\lambda+1=0$

$(\lambda-1)^2=0 \quad \lambda_{1,2}=1$

$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \xrightarrow{R_1+R_2} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ v_1: leading $\quad m(\lambda=1)=1 < \text{alg}(\lambda=1)=2$

v_2: free.

Kuchak $v^{(2)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad v^{(1)} = (A-I)v^{(2)} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$P = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \quad J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

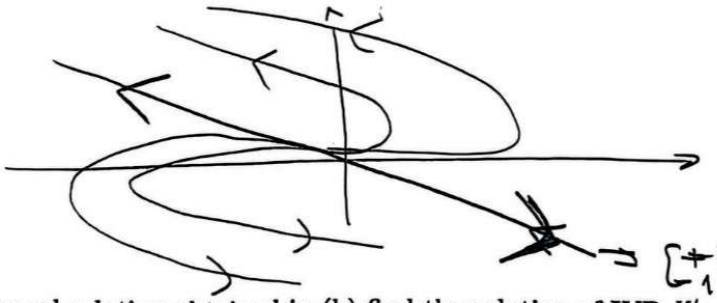
(b) Find general solution of the system $X' = AX$ using $\Xi(t) = Pe^{Jt}$.

$X(t) = P \cdot e^{Jt}$

$X_{general} = \underbrace{\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}}_{P} \cdot \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = c_1 e^t \begin{bmatrix} 1 \\ -1 \end{bmatrix} +$

$c_2 e^t \left(t\begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} \cdot \end{bmatrix} \right)$ ($v^{(1)}$, $v^{(2)}$)

(c) Draw phase portrait.



(c) Using general solution obtained in (b) find the solution of IVP $X' = AX$, $X(0) = (1,2)^T$

$X_{general}(0) = P\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_1+c_2 \\ -c_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$X_{general}(t) = P \cdot e^{Jt}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$

$c_1+c_2=1 \Rightarrow c_2=3$

$-c_1=2 \Rightarrow c_1=-2.$

$X(t) = 3e^t\begin{bmatrix} t+1 \\ -t \end{bmatrix} - 2e^t\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} e^t+3te^t \\ -3te^t+2e^t \end{bmatrix}$

DERYA ALTINTAN ♡ GOOD LUCK ♡

$\ker(A-I) = \text{span}\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} : v_1+v_2=0\right) = \text{span}\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = V_{\lambda=1}$

$(A-I)^2\vec{v} = 0 \quad \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $v^{(2)} \in \ker(A-I)$ $\notin \ker(A-I)^2$

$\ker(A-I)^2 = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\} \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$